Seat No.

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 2

Subject: 394172 Mathematics II

Date: 19 April 2019

Year: 2018

Section: 15-18

Time: 13.00-16.00

Name:____________________ ID:____________________ Class:__________

Directions: The test is designed to measure your comprehension. The test is divided into 10 questions. There will be 9 pages (including this page) and they are worth 100 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Question 1. Consider the following system of equations

$$\begin{aligned} 2x - ay &= 5, \\ 3x + 7y &= -4, \end{aligned}$$

where a is a real number.

1.1 If $a=3$, then find the solution of the above system. (7 points)

1.2 Find the value of a such that the above system has no solution. (3 points)

Question 2. Solving the system of three nonlinear equations. (10 points)

$$x-y-z=1,$$
$$x^2-(y-z)^2=21,$$
$$(x-y)^2-z^2=3.$$

Question 3. 3.1 If $\left(9+\sqrt{6}+\sqrt{10-4\sqrt{6}}\right)^{\frac{1}{2}} = \sqrt{a}+b$, where $a,b \in R$, then find the value of $\dfrac{5}{\sqrt{a}-\sqrt{b}}$. (5 points)

3.2 Solve the radical equation $\sqrt{x+7}+\sqrt{x+2}=\sqrt{6x+13}$. (5 points)

Question 4. To find the solution set of the exponential equation

$$\frac{27^{-3x}\left(\frac{1}{9}\right)^{2x-1}}{81^{2}} = \frac{1}{243}. \quad \text{(10 points)}$$

Question 5. Given the exponential inequality $\sqrt{9^{x}-3^{x+2}} > 3^{x}-9$. Find the solution set. (10 points)

Question 6. Given $\frac{1}{4}\log_a \frac{81}{16} - 4\log_{\sqrt{3}} 2 + \frac{1}{3}\log_{\sqrt{3}} 216 + \log_{\sqrt{3}} \frac{3}{a} + 2\log_{\sqrt{3}} 5 + 6\log_{\sqrt{3}} 2 = 6$.
Evaluate the value of $\sqrt{a}$. (10 points)

Question 7. Find all solutions of the logarithmic equation

$$(\log 100x)^2 + 2\log_{100} x + 2 = 0.$$ (10 points)

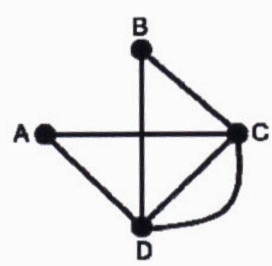
Question 8. Let $a, x,$ and y be integers.

8.1 Determine whether the statement " If $a \mid (2x+4y)$, then $a \mid (2x)$ and $a \mid (4y)$ " is correct or not. Give an example to support your answer. (5 points)

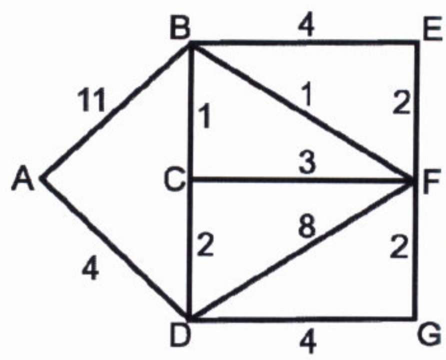
8.2 If $a^6 \mid 3x - 3y$ and $a \mid 5x + y$, show that $a \mid 36x$. (5 points)

Question 9. Find gcd (244,354) using Euclidean algorithm and then find lcm (244,354). (10 points)

Question 10. 10.1 Determine whether the following graph has an Euler circuit. If it has, find an Euler circuit of this graph. (5 points)



10.2 Find the shortest path between the vertex A and the vertex E in the weighted graph. (5 points)



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